

Day 92: Descriptive Statistics

1. Day number

Day 92

2. Topic name

Mean, Median, Variance, and Standard Deviation

3. Connection

You can already represent numerical datasets using Python lists and NumPy arrays.

Today you will learn how to **summarize how those numbers behave**.

Instead of looking at every individual value, descriptive statistics can answer questions such as:

- What is the typical value?
 - Where is the middle of the data?
 - Are the values close together?
 - Are the values spread far apart?
-

4. Important topics

The main ideas are:

- **Mean** — the average value
- **Median** — the middle value
- **Spread** — how far apart the values are
- **Variance** — average squared distance from the mean
- **Standard deviation** — a more intuitive measurement of spread

A useful mental model is:

Mean/median → **center of the data**

Variance/standard deviation → **spread of the data**

5. Foundational notes

Consider monthly sales:

10, 12, 14, 16, 18

There are two basic things we might want to know.

First, what value represents the **center**?

Second, how much do the individual values **move away from that center**?

Descriptive statistics help answer these questions without making predictions about the future.

6. Understanding each concept intuitively

Mean

The **mean** is what we normally call the **average**.

Imagine combining all the sales and redistributing them equally across the five months. The amount each month would receive is the mean.

Conceptually:

Mean = total of all values / number of values

For:

10, 12, 14, 16, 18

the total is:

70

There are five values:

$70 / 5 = 14$

So:

Mean = 14

Median

The **median** is the value sitting in the middle after the numbers have been sorted.

10, 12, 14, 16, 18
 ↑
 middle

Therefore:

Median = 14

For an odd number of values, there is one middle value.

For an even number of values, we normally take the average of the two middle values.

Spread

Two datasets can have the same mean but behave very differently.

For example:

Dataset A: 13, 14, 14, 14, 15
Dataset B: 2, 8, 14, 20, 26

Both are centered around 14, but Dataset B is much more spread out.

That is why the mean alone does not completely describe a dataset.

Variance

Variance measures how far the values are spread around the mean.

The basic idea is:

1. Find the mean.
2. Measure how far every value is from the mean.
3. Square those distances.
4. Find their average.

Squaring prevents positive and negative differences from cancelling each other.

For our example, the mean is 14.

Value	Difference from mean	Squared difference
10	-4	16
12	-2	4
14	0	0
16	2	4
18	4	16

The squared differences total:

$$16 + 4 + 0 + 4 + 16 = 40$$

For this beginner example, divide by the number of values:

$$40 / 5 = 8$$

So:

Variance = 8

A **small variance** means the numbers are relatively close together.

A **large variance** means they are more spread out.

Standard deviation

Standard deviation is closely related to variance.

It is simply the **square root of the variance**.

For our example:

Variance = 8

Standard deviation = $\sqrt{8}$
 ≈ 2.83

So:

Standard deviation ≈ 2.83

Standard deviation is often easier to interpret because it uses roughly the **same units as the original data**.

For example, if sales were measured in thousands of units, the standard deviation would also be interpreted in thousands of units.

7. Easy numerical example

Suppose five months produced these sales:

10, 12, 14, 16, 18

The main descriptive statistics are:

Statistic	Result
Mean	14
Median	14
Variance	8
Standard deviation	≈ 2.83

You can read this as:

Sales are centered around 14, and the individual monthly values have some moderate spread around that center.

8. Problem statement

Given these five monthly sales numbers:

20, 25, 25, 30, 50

Your task is to:

1. Calculate the **mean**.
2. Calculate the **median**.
3. Think about how far the numbers are from the mean.

4. Explain what **variance** tells you about these sales.
5. Explain what **standard deviation** tells you about these sales.

Notice that 50 is quite far from the other values. Think about how this might affect the mean and the spread.

9. Concepts used

You will use:

- numerical data
- sorting
- addition
- division
- mean
- median
- difference from the mean
- squaring
- variance
- square root
- standard deviation

These ideas will become important later when working with **data analysis and machine learning**.

10. Thought process

When you receive a small numerical dataset, think in this order:

```
What are my values?
  ↓
What is their center?
  ↓
Calculate mean
  ↓
Sort them and find median
  ↓
How far is each value from the mean?
  ↓
Square those differences
  ↓
Calculate variance
  ↓
Square root the variance
  ↓
Get standard deviation
```

Do not try to memorize everything at once.

Remember the simpler distinction:

Mean + Median

↓

Where is the data centered?

Variance + Standard deviation

↓

How spread out is the data?

11. Beginner-friendly calculation steps

Suppose:

values = 10, 12, 14, 16, 18

Step 1: Add the values

$10 + 12 + 14 + 16 + 18$
 $= 70$

Step 2: Calculate the mean

$70 / 5$
 $= 14$

Step 3: Find the median

The values are already sorted:

10, 12, 14, 16, 18

Middle value:

14

Step 4: Find differences from the mean

$10 - 14 = -4$
 $12 - 14 = -2$
 $14 - 14 = 0$
 $16 - 14 = 2$
 $18 - 14 = 4$

Step 5: Square the differences

$$\begin{aligned}(-4)^2 &= 16 \\ (-2)^2 &= 4 \\ 0^2 &= 0 \\ 2^2 &= 4 \\ 4^2 &= 16\end{aligned}$$

Step 6: Calculate variance

$$\begin{aligned}(16 + 4 + 0 + 4 + 16) / 5 \\ = 40 / 5 \\ = 8\end{aligned}$$

Step 7: Calculate standard deviation

$$\sqrt{8} \approx 2.83$$

12. How Python/NumPy could calculate them conceptually

NumPy already provides functions for these calculations.

Conceptually:

```
import numpy as np

sales = np.array([...])

mean_value = np.mean(sales)
median_value = np.median(sales)
variance_value = np.var(sales)
standard_deviation = np.std(sales)
```

Think of the functions as:

```
np.mean()
  ↓
Find the average

np.median()
  ↓
Find the middle

np.var()
  ↓
Measure squared spread
```

```
np.std()
```



Measure spread in a more intuitive form

For now, understanding **what each result means** is more important than memorizing the functions.

13. Easy edge cases

All values are equal

Suppose:

```
20, 20, 20, 20, 20
```

Mean:

```
20
```

Every value is exactly equal to the mean.

Therefore there is no spread:

```
Variance = 0  
Standard deviation = 0
```

This makes intuitive sense: nothing varies.

Only one value

Suppose:

```
25
```

Then:

```
Mean = 25  
Median = 25
```

For the simple population-style calculation used here:

```
Variance = 0  
Standard deviation = 0
```

There is only one value, so there is no spread between values.

14. Expected calculations

For the example:

10, 12, 14, 16, 18

you should obtain:

Mean = 14

Median = 14

Squared differences:

16, 4, 0, 4, 16

Variance = 8

Standard deviation ≈ 2.83

The key interpretation is:

14 describes the center.

8 and 2.83 describe the spread around that center.

One important beginner distinction is:

Mean → average

Median → middle

Variance → squared spread

Std. dev. → easier-to-interpret spread

15. Hint only

For your exercise:

20, 25, 25, 30, 50

Start by calculating:

$20 + 25 + 25 + 30 + 50$

Then divide by the **number of months**.

For the median, sort the numbers and look for the **middle value**.

For variance, first ask:

| How far is each sales number from the mean?

For standard deviation, remember:

standard deviation = square root of variance

Also pay special attention to . Ask yourself: **does a value far away from the others increase or decrease the spread?**